

Rethinking Investments in Human Capital in the Age of Artificial Intelligence

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Abstract

How does Artificial Intelligence reshape human capital investment decisions? I develop an Overlapping Generations model where risk-averse agents choose training paths based on AI's expected impact and profound uncertainty. Calibrated to the US labor market, the framework decomposes AI shocks into expected automation, expected augmentation, and their associated uncertainties. The model yields three key results. First, both expected automation and its uncertainty drive a structural reallocation, causing workers to flee highly uncertain cognitive professions toward manual positions. Second, expected wages rise across all paths (+12% to +20%). Third, automation uncertainty generates an endogenous risk premium.

1 Introduction

For most of the twentieth century, technological progress has pointed in the direction of incentivizing investments in human capital. Successive waves of automation primarily displaced routine, mechanical tasks concentrated in low and middle-skill occupations, while simultaneously raising the demand for high-skill workers with higher education (Acemoglu and Autor, 2011; Autor et al., 2003). In this environment, a college degree has been a relatively safe investment, offering high expected earnings and low unemployment risk. The secular rise in the college wage premium and in educational attainment throughout the latter half of the twentieth century is largely a product of this dynamic (Autor, 2019).

Artificial Intelligence (AI) fundamentally disrupts this long-standing relationship. Unlike previous technologies, generative AI demonstrates a high degree of proficiency precisely in the non-routine, cognitive tasks that have historically been the exclusive domain of high-skill workers (Eloundou et al., 2024; Webb, 2019; Tomlinson et al.,

2025; Massenkoff and McCrory, 2026). The implications are potentially far-reaching: the very occupations that attracted generations of students to invest heavily in human capital are now among the most exposed to this new technology.

A critical feature of this technological breakthrough, however, is its profound *uncertainty*. The net effect of AI on labor demand is not only unknown, but its expected impact and the magnitude of the uncertainty vary dramatically across occupations. The nature of this uncertainty is due to the fact that it is very difficult to predict exactly which tasks will be automated and which will be augmented. Furthermore, any technology which can substitute or augment human labor at the task level has an ambiguous impact on wages. As is standard in a task-based model, when a task is automated, two opposing effects arise: a displacement effect that acts as a negative force by automating tasks, depressing labor demand, and a productivity effect that raises demand for tasks that are not automated (Acemoglu and Restrepo, 2018). Similarly, when a task is augmented (labor productivity increases at the task level), the worker is more productive, but the price of what he produces decreases because of the downward sloping demand curve (Althoff and Reichardt, 2026). In both cases, which of the two effects will eventually dominate in each occupation is unknown, but individuals must make irreversible decisions regarding their human capital investments today. The key point of this paper is that individuals do not simply choose based on expectations about the future, but they also take into account the uncertainty surrounding those expectations. For example, manual jobs have relatively few tasks exposed to potential automation, hence uncertainty about the future is lower. On the other hand, training paths like legal professions and computer science face high uncertainty about the future, as many tasks in their occupations could be either automated or significantly augmented.

To formalize and quantify these dynamics, I develop an Overlapping Generations (OLG) model in which heterogeneous, risk-averse agents choose among J training paths to maximize their discounted expected utility of future wages, net of idiosyncratic investment costs. The key innovation is the decomposition of the AI shock for each training path j into four distinct components: an expected automation effect $\mu_{A,j}$, an expected augmentation effect $\mu_{\gamma,j}$, and their respective variance components $\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$, which capture the uncertainty about AI’s ultimate capabilities. Agents are risk averse and invest in human capital to maximize their expected utility. The individual discrete choice problem aggregates to a tractable multinomial logit equilibrium.

I calibrate the model to the US labor market. The task-level AI shocks are derived from Althoff and Reichardt (2026), who computed for each task whether it is theoretically automatable with generative AI and an expected augmentation factor if instead generative AI acts as an augmentation technology in that task. Moreover, I rely on the current AI task penetration (Massenkoff and McCrory, 2026), which measures the

current AI usage in each task according to Anthropic data. The employment shares across 14 training paths are constructed from the IPUMS American Community Survey (ACS) for 2022, while the median wages are collected from the Bureau of Labor Statistics (BLS). I recover the structural training cost parameters by inverting the pre-AI steady-state equilibrium (2022), and simulate the resulting post-AI equilibrium.

The model delivers three main results, linking theoretical mechanisms to their quantitative implications for the US economy. First, both expected automation and its profound uncertainty drive a structural reallocation of labor. Because agents are risk-averse, an increase in AI-driven uncertainty unambiguously reduces equilibrium employment in exposed training paths, prompting individuals to systematically flee highly volatile cognitive professions and seek refuge in safer manual positions. By contrast, expected augmentation and its associated uncertainty are not major drivers of labor reallocation. While still economically important, these components of the AI shock are largely uniform across training paths. Second, the model predicts an across-the-board increase in expected wages (ranging from +12% to +20%). Importantly, this wage growth is heavily mediated by labor market dynamics: when a safe-haven training path attracts an influx of workers, it creates congestion that erodes marginal productivity and dampens individual wage gains. Third, if we ignore uncertainty about future automation, the impact on expected wages is very similar across all training paths, approximately +13/14%. However, when we add uncertainty about future automation, it uniquely generates an endogenous risk premium. This is a pure general equilibrium effect rather than the standard risk premium arising from investment in a risky asset: the massive flight of risk-averse workers from uncertain cognitive paths reduces labor supply in those sectors, creating a labor scarcity that mechanically pushes their expected wages upward, even in the absence of any direct productivity gain. Similarly, the inflow of workers in safer training paths creates a congestion of workers that put downward pressure on wages in those training paths.

These three mechanisms interact in ways that are difficult to detect using reduced-form methods alone. A highly exposed cognitive path may see its expected wage rise purely due to the endogenous risk premium, even if AI has a mildly negative expected productivity impact. Conversely, a safe-haven path may experience disappointing wage growth due to labor market congestion. This work provides a structural framework to disentangle these intertwined effects.

This paper contributes to three strands of the literature. First, it builds on the large empirical literature on AI’s impact on the labor market (among many others, Acemoglu and Restrepo 2018; Webb 2019; Agrawal et al. 2025; Ide and Talamàs 2025; Acemoglu et al. 2025; Gans and Goldfarb 2026), by providing a structural model that allows for a clean decomposition of the supply-side response to the demand and uncertainty

channels. Second, it contributes to the literature on human capital investment under uncertainty (Levhari and Weiss, 1974; Cunha et al., 2005; Cunha and Heckman, 2016), by introducing a new form of risk in the context of human capital investment decisions. Third, it connects to the macro literature on skill-biased technological change (Acemoglu and Autor, 2011; Autor et al., 2003; Autor, 2019) by characterizing the novel mechanism through which AI reshapes economic incentives in ways that are qualitatively different from prior automation waves.

In this regard, the closest paper to this work is Althoff and Reichardt (2026), which provides the foundational task-level framework for how AI simultaneously automates, augments, and simplifies human labor. While they embed these mechanisms into a rich dynamic model of occupational choice, their analysis evaluates the labor market transition under the assumption that uncertainty is symmetric across occupations. Furthermore, they treat initial skills as exogenous and focus on skill accumulation on the job. This paper takes a complementary approach by explicitly modeling the training-specific uncertainty surrounding AI’s ultimate capabilities and focusing on the endogenous pre-labor market training choice. By introducing risk-averse agents who must make irreversible human capital investments before entering the labor market, I abstract from on-the-job skill accumulation to isolate a novel supply-side mechanism driven by training-specific uncertainty. This allows me to quantify how technological uncertainty itself, distinct from expected productivity changes, drives labor reallocation, creates congestion in safe-haven occupations, and generates endogenous risk premiums in highly uncertain occupations.

The rest of the paper is organized as follows. Section 2 develops the theoretical model. Section 3 describes the measurement of the AI shock. Section 4 describes the calibration strategy, and Section 5 presents the quantitative simulation results. Finally, Section 6 concludes.

2 The Model

I consider an Overlapping Generations (OLG) economy where agents live for two periods: young ($g = y$) and old ($g = o$). The economy features a finite set of potential training paths. Each training path involves a period of training (when young) followed by a period of trained employment (when old). The only exception is the case of no training, which represents a fallback option with no investment in human capital. Training paths may be of two types: a *technical* training path (e.g., plumber, cook) is characterized by a period of apprenticeship as training, while a *cognitive* training path involves a period of studying in college in a certain field before entering the labor market. Let’s denote the different training paths by $j \in \{0, 1, \dots, J\}$ ($j = 0$ refers to the

case of no training). When young ($g = y$), agents are endowed with a unit of time and choose a training path j . This choice involves an investment in human capital with uncertain returns. In the second period ($g = o$), agents supply labor inelastically, realize their wage (which depends on the chosen training path and labor market conditions), and consume.

2.1 Environment

A representative firm demands labor for each training path. The output per worker for training j is

$$Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} y_{\tau}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}}$$

where $\mathcal{T}_j$ is the set of relevant tasks for training j , y_{τ} is the output of task τ , and $\theta_{j,\tau}$ is the importance weight of task τ in the production for training j .

The production function for task τ is

$$y_{\tau}(l_{\tau}, k_{\tau}) = \begin{cases} l_{\tau} \gamma_{\tau} & \text{if } \tau \in \mathcal{N}_j \\ l_{\tau} \gamma_{\tau} + \phi_{\tau} k_{\tau}, & \text{if } \tau \in \mathcal{A}_j \end{cases}$$

where $\mathcal{A}_j$ and $\mathcal{N}_j$ are, respectively, the sets of automatable and non-automatable tasks in training j , l_{τ} is the share of time a worker with training j spends in task τ , and k_{τ} is the capital per worker in task τ . Finally, γ_{τ} and ϕ_{τ} are parameters that capture the productivity of labor and capital in task τ . In this framework, automation refers to an expansion of the set $\mathcal{A}_j$, and labor augmentation refers to an increase in γ_{τ} .

The problem solved by the representative firm is

$$\begin{aligned} \max_{l_{j,\tau}, k_{j,\tau}} \quad & L_j \left(p_j Y_j - w_j - R \sum_{\tau \in \mathcal{A}_j} k_{j,\tau} \right) \\ \text{subject to } \quad & Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \tilde{y}_{\tau}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \\ & \tilde{y}_{\tau} = y_{\tau}(l_{\tau}, k_{\tau}) \\ & \sum_{\tau \in \mathcal{T}_j} l_{j,\tau} = 1 \end{aligned}$$

where L_j is the number of workers with training j in the labor market. Importantly, L_j is fixed from previous period choice and it is not possible to increase it or decrease

it until next period (or next generation).

Assuming that all automatable tasks are produced with capital, i.e., $\frac{w_j}{\gamma_\tau} > \frac{R}{\phi_\tau}$ for all $\tau \in \mathcal{A}_j$, the labor demand for task τ is zero for $\tau \in \mathcal{A}_j$. Thus, the labor demand is given by

$$l_{j,\tau} = \frac{\theta_{j,\tau} \gamma_\tau^{\rho-1}}{\sum_{\tau' \in \mathcal{N}_j} \theta_{j,\tau'} \gamma_{\tau'}^{\rho-1}}$$

and capital per worker is

$$k_{j,\tau} = \theta_{j,\tau} \phi_\tau^{\rho-1} Y_j \left(\frac{p_j}{R} \right)^\rho$$

As a result the optimal production per worker in training j is

$$Y_j = \Gamma_j^{\frac{\rho}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}}$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$ is the labor share for training j .

Finally, since markets are perfectly competitive, the wage is given by

$$w_j = p_j Y_j \Gamma_j = p_j \Gamma_j^{\frac{1}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}} \quad (1)$$

2.2 Equilibrium

Following Althoff and Reichardt (2026), the price of each occupational good p_j is determined in equilibrium through demand and supply. Labor supply is fixed by the previous period's training choice. The workers, in turn, consume and generate demand for the occupational goods. I assume that demand is characterized by a homothetic and invertible demand function $D_j(\{p_j\}_{j=0}^J)$ that maps prices p_j into relative demand for each occupational good.

Definition (Competitive Equilibrium): Given a distribution of workers across training paths $\{L_j\}_{j=0}^J$ such that $\sum_{j=0}^J L_j = 1$, and the supply of capital $\mathcal{K}$, a competitive equilibrium in this economy is given by a sequence of prices $\{p_0, p_1, \dots, p_J, R\}$, such that:

1. Demand for goods equals supply: $D_j(\{p_j\}_{j=0}^J) = L_j Y_j$
2. Demand for capital equals supply: $\mathcal{K} = \sum_{j=0}^J \sum_{\tau \in \mathcal{A}_j} L_j k_{j,\tau}$

2.3 Training Choice

Agents are risk-averse, with preferences described by a CRRA utility function $u(w) = \frac{w^{1-\eta}}{1-\eta}$. In what follows, I will use the log wage, so the equivalent utility function over the log wage is $u(\ln w) = \frac{e^{(1-\eta)\ln w}}{1-\eta}$. Individuals are heterogeneous with respect to a vector of net costs $C_i = [C_{i0}, C_{i1}, \dots, C_{iJ}]$, which aggregates all non-pecuniary disutility and direct expenses specific to each training path. For example, if a worker really hates math, the cost of choosing engineering will be huge, whereas if she really loves literature, the cost of choosing humanities will be very low.

A young agent i chooses the training path j that maximizes the discounted expected utility of future earnings, net of the investment cost:

$$V_i^y(C_i) = \max_j \{ \beta \mathbb{E}_t [u(\ln w_j)] - C_{ij} \} \quad (2)$$

where $\beta \in (0, 1)$ is the discount factor and the log-wage for training path j is given by Equation 1:

$$\ln w_j = \ln p_j + \frac{1}{1-\rho} \ln \Gamma_j(\mathcal{A}_j) + \frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$. As in other standard task-based models, automation has an ambiguous impact because it shrinks the set of tasks performed by humans (displacement effect), but it increases productivity in the remaining tasks due to the cost-savings in the automated tasks (productivity effect); also, it generates a change in the equilibrium prices, which has a direct impact on wages. Similarly, augmentation has an ambiguous impact on wages because it increases productivity of labor, but it changes the equilibrium prices. However, when young, agents do not know what will be the final set of automatable tasks $\mathcal{A}_j$ in each training j , nor which tasks will be augmented. To make this mathematically tractable, I map the set of tasks into a continuum and define a continuous scalar $A_j \in [0, 1]$ representing the share of automated tasks (weighted by importance), and a scalar γ_j representing the average augmentation factor across tasks. Let us suppose their beliefs about the future automation share A_j are well proxied by a normal distribution, $\mathcal{N}(\mu_{A,j}, \sigma_{A,j}^2)$, where $\mu_{A,j}$ is the future expected automation share for training j and $\sigma_{A,j}^2$ captures the uncertainty surrounding future automation in training j . Similarly, for augmentation, workers' beliefs about γ_j are proxied by a normal distribution $\mathcal{N}(\mu_{\gamma,j}, \sigma_{\gamma,j}^2)$, where $\mu_{\gamma,j}$ is the expected augmentation for training j and $\sigma_{\gamma,j}^2$ captures the uncertainty around it. Importantly, all the parameters capturing beliefs are training-specific. Thus, there are some training paths characterized

by higher risk of automation and others where augmentation is more likely; also, some training paths could be relatively less subject to both automation and augmentation, but they still can be indirectly impacted through demand spillovers. Moreover, the variance is also training specific and it captures the fact that some training paths are fundamentally more uncertain than others. In the empirical part, I use the existing evidence to calibrate the two moments for each distribution.

Because the log-wage $\ln w_j$ is a non-linear function of both the automated task share A_j and the augmentation factor γ_j , its exact distribution is analytically intractable. To restore tractability, I rely on a first-order Taylor expansion (the Delta method) around the expected shocks. Assuming the automation and augmentation shocks are independent, the linearized log-wage is approximately normally distributed, with expected future log-wage

$$\mathbb{E}[\ln w_j] = \ln p_j(\{L_k, \mu_{A,k}, \mu_{\gamma,k}\}_{k=0}^J) + \frac{1}{1-\rho} \ln \Gamma_j(\mu_{A,j}) + \frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \mu_{\gamma,j}^{\rho-1} \right)$$

where the endogenous expected price depends on the reallocation of labor supply, and expected automation and augmentation; the expected labor share in each training j depends on expected automation; finally, the last term captures the expected productivity, which depends on expected automation and on the share of tasks that will not be automated. The overall variance is the sum of the variances of the underlying shocks, weighted by the squared sensitivities:

$$Var(\ln w_j) = \left(\frac{\partial \ln w_j}{\partial A_j} \bigg|_{\mu_{A,j}, \mu_{\gamma,j}} \right)^2 \sigma_{A,j}^2 + \left(\frac{\partial \ln w_j}{\partial \gamma_j} \bigg|_{\mu_{A,j}, \mu_{\gamma,j}} \right)^2 \sigma_{\gamma,j}^2$$

Given this normal approximation for the log-wage, the expected utility can be evaluated exactly using the moment-generating function of the normal distribution:

$$\mathbb{E}[u(\ln w_j)] = \frac{1}{1-\eta} \exp \left((1-\eta) \mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j) \right)$$

The certainty equivalent Ω_j is defined as the guaranteed wage that yields the same expected utility, $u(\Omega_j) = \mathbb{E}[u(\ln w_j)]$. Equating the two yields:

$$\frac{e^{(1-\eta)\Omega_j}}{1-\eta} = \frac{1}{1-\eta} \exp \left((1-\eta) \mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j) \right)$$

Canceling the constant term, taking logs, and rearranging provides the exact certainty

equivalent for training path j :

$$\Omega_j = \underbrace{\ln p_j + \frac{1}{1-\rho} \ln \Gamma_j + \frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \mu_{\gamma,j}^{\rho-1} \right)}_{\mathbb{E}(\ln w_j)} - \underbrace{\frac{\eta-1}{2} \text{Var}(\ln w_j)}_{\text{Risk premium}}$$

which formalizes exactly how agents penalize uncertainty depending on their degree of risk aversion η . The main novelty of this model is that it creates a training-specific risk premium that is linked with the relative uncertainty regarding future automation and augmentation due to AI in each training path.

Substituting $\mathbb{E}[u(\ln w_j)]$ with $u(\Omega_j)$, the agent's objective function becomes:

$$V_i^y(C_i) = \max_j \{ \beta u(\Omega_j) - C_{ij} \}$$

To render the discrete choice problem analytically tractable, I apply a first-order Taylor expansion to the utility of the certainty equivalent $u(\Omega_j)$ around a baseline average log-wage $\bar{\mu}$. Substituting $u(\Omega_j) \approx u(\bar{\mu}) + u'(\bar{\mu})(\Omega_j - \bar{\mu})$ into the objective, the maximization problem simplifies to:

$$\max_j \{ \beta \Omega_j - C_{ij}^* \} \quad (3)$$

where $C_{ij}^* \equiv C_{ij}/u'(\bar{\mu})$ is the normalized cost parameter. Notice that, although we are linearizing the utility function the risk aversion coefficient η still enters into the decision rule through the risk premium in the certainty equivalent Ω_j .

I assume that the normalized cost is determined by a training-specific common average cost, $\bar{C}_j^*$, and an idiosyncratic preference component, such that $C_{ij}^* = \bar{C}_j^* - \epsilon_{ij}$. Assuming the idiosyncratic shocks ϵ_{ij} are independently and identically distributed following a Type I Extreme Value (Gumbel) distribution with location parameter zero and scale parameter ζ , the aggregate market share of labor choosing training path j is mapped by the standard multinomial conditional logit formulation (McFadden (1972)):

$$L_j^* = \frac{\exp((\beta \Omega_j - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta \Omega_h - \bar{C}_h^*)/\zeta)} \quad (4)$$

2.4 Theoretical Results and Equilibrium Properties

Using the structural discrete choice framework derived above, this section establishes the comparative statics of the model with respect to the four parameters of the agents' beliefs about the automation and augmentation shocks: the expected levels ($\mu_{A,j}$ and $\mu_{\gamma,j}$) and their respective uncertainties ($\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$).

Because output per worker Y_j is independent of L_j , an increase in labor supply L_j mechanically increases the aggregate supply of occupational good j . Given the homothetic demand function, this lowers the equilibrium price p_j , implying a downward-sloping labor demand curve: $\frac{\partial \ln w_j}{\partial L_j} = \frac{\partial \ln p_j}{\partial L_j} < 0$. This allows us to map how exogenous changes in beliefs affect actual equilibrium wages. Note that the derivatives derived from the conditional logit formula in Propositions 1 and 5 represent the *direct* (or partial equilibrium) shifts in labor supply holding expected wages and prices constant. In general equilibrium, these initial shifts trigger price adjustments that ultimately dampen the final reallocation; however, this effect can never reverse the direction of the response.

Proposition 1 (Flight from Uncertainty). *An increase in automation uncertainty ($\uparrow \sigma_{A,j}^2$) or augmentation uncertainty ($\uparrow \sigma_{\gamma,j}^2$) for training path j leads to a decrease in the equilibrium employment L_j^* :*

$$\frac{\partial L_j^*}{\partial \sigma_{A,j}^2} = -\frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial A_j} \right)^2 L_j^*(1-L_j^*) < 0 \quad (5)$$

$$\frac{\partial L_j^*}{\partial \sigma_{\gamma,j}^2} = -\frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial \gamma_j} \right)^2 L_j^*(1-L_j^*) < 0 \quad (6)$$

Intuition: As a career path becomes more unpredictable due to exposure to AI, risk-averse individuals systematically flee to safer alternatives. The magnitude of this structural flight depends directly on their degree of risk aversion (η).

Proposition 2 (Endogenous Risk Premium). *An increase in either automation or augmentation uncertainty for training path j ($\uparrow \sigma_{A,j}^2, \uparrow \sigma_{\gamma,j}^2$) raises the expected wages $\mathbb{E}(\ln w_j)$. This reflects an endogenous risk premium driven by labor scarcity:*

$$\frac{\partial \mathbb{E}(\ln w_j)}{\partial \sigma_{A,j}^2} = \underbrace{\frac{\partial \mathbb{E}(\ln w_j)}{\partial p_j}}_{(>0)} \underbrace{\frac{\partial p_j}{\partial L_j}}_{(<0)} \underbrace{\frac{\partial L_j^*}{\partial \sigma_{A,j}^2}}_{(<0)} > 0, \quad \frac{\partial \mathbb{E}(\ln w_j)}{\partial \sigma_{\gamma,j}^2} = \underbrace{\frac{\partial \mathbb{E}(\ln w_j)}{\partial p_j}}_{(>0)} \underbrace{\frac{\partial p_j}{\partial L_j}}_{(<0)} \underbrace{\frac{\partial L_j^*}{\partial \sigma_{\gamma,j}^2}}_{(<0)} > 0 \quad (7)$$

Intuition: This characterizes one of the core theoretical mechanisms of the model. Uncertainty reduces the share of labor choosing training path j (see proposition 1). The resulting labor scarcity pushes up the equilibrium price of the occupational good, mechanically increasing the realized wage for those who remain.

Proposition 3 (Expected Shocks and Labor Supply). *Because both expected automation ($\mu_{A,j}$) and expected augmentation ($\mu_{\gamma,j}$) have theoretically ambiguous direct effects on the certainty equivalent Ω_j (due to displacement, productivity, and price effects), their impact on equilibrium employment L_j^* , $\frac{\partial L_j^*}{\partial \mu_{A,j}}$ and $\frac{\partial L_j^*}{\partial \mu_{\gamma,j}}$ is also theoretically ambiguous.*

Proposition 4 (Expected Wages and Crowding Effect). *An increase in expected augmentation ($\uparrow \mu_{\gamma,j}$) or a change in expected automation ($\mu_{A,j}$) directly shifts the expected log-wage. However, the subsequent endogenous reallocation of labor (L_j^*) generates a crowding effect that dampens the final realized expected wage response:*

$$\frac{dE[\ln w_j]^*}{d\mu_{\gamma,j}} = \underbrace{\frac{\partial}{\partial \mu_{\gamma,j}} \left[\frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \mu_{\gamma,j}^{\rho-1} \right) \right]}_{\text{direct effect } (>0)} + \underbrace{\frac{\partial \ln p_j}{\partial \mu_{\gamma,j}}}_{\text{direct price effect } (<0)} + \underbrace{\frac{\partial \ln p_j}{\partial L_j} \frac{dL_j^*}{d\mu_{\gamma,j}}}_{\substack{(<0) \\ (\text{ambig.})}} \quad (8)$$

$$\begin{aligned} \frac{dE[\ln w_j]^*}{d\mu_{A,j}} &= \underbrace{\frac{\partial}{\partial \mu_{A,j}} \left[\frac{1}{1-\rho} \ln \Gamma_j(\mu_{A,j}) \right]}_{\text{displacement effect } (<0)} + \underbrace{\frac{\partial}{\partial \mu_{A,j}} \left[\frac{1}{\rho-1} \ln \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \mu_{\gamma,j}^{\rho-1} \right) \right]}_{\text{productivity effect } (>0)} \\ &+ \underbrace{\frac{\partial \ln p_j}{\partial \mu_{A,j}}}_{\text{direct price effect } (<0)} + \underbrace{\frac{\partial \ln p_j}{\partial L_j} \frac{dL_j^*}{d\mu_{A,j}}}_{\substack{(<0) \\ (\text{ambig.})}} \end{aligned} \quad (9)$$

Because the labor demand curve is downward sloping ($\frac{\partial \ln p_j}{\partial L_j} < 0$), the influx (or outflux) of workers offsets part of the initial productivity and direct price shifts, making the net expected wage response strictly smaller in magnitude than the direct effect.

Proposition 5 (Cross-Career Spillovers - Employment). *A shock that increases the certainty equivalent Ω_j for training path j (which depends ambiguously on $\mu_{A,j}, \mu_{\gamma,j}$, and negatively on $\sigma_{A,j}^2, \sigma_{\gamma,j}^2$) leads to a decrease in the employment level of all other careers $h \neq j$:*

$$\frac{\partial L_h^*}{\partial \mu_{A,j}} = -\frac{\beta}{\zeta} \frac{\partial \Omega_j}{\partial \mu_{A,j}} L_h^* L_j^*, \quad \frac{\partial L_h^*}{\partial \sigma_{A,j}^2} = \frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial A_j} \right)^2 L_h^* L_j^* > 0 \quad (10)$$

$$\frac{\partial L_h^*}{\partial \mu_{\gamma,j}} = -\frac{\beta}{\zeta} \frac{\partial \Omega_j}{\partial \mu_{\gamma,j}} L_h^* L_j^*, \quad \frac{\partial L_h^*}{\partial \sigma_{\gamma,j}^2} = \frac{\beta(\eta-1)}{2\zeta} \left(\frac{\partial \ln w_j}{\partial \gamma_j} \right)^2 L_h^* L_j^* > 0 \quad (11)$$

Intuition: Human capital supply is a zero-sum game. If AI makes training path j more attractive, it inevitably drains workers from other paths h . Conversely, if AI makes career j less attractive, workers actively seek refuge in relatively positively affected careers.

Proposition 6 (Cross-Career Spillovers - Realized Wages). *A shock that increases the certainty equivalent Ω_j for training path j (which depends ambiguously on $\mu_{A,j}, \mu_{\gamma,j}$, and negatively on $\sigma_{A,j}^2, \sigma_{\gamma,j}^2$) affects the expected wages of all other careers $h \neq j$ through*

two general equilibrium channels: labor reallocation and aggregate demand spillovers:

$$\frac{d\mathbb{E}[\ln w_h]^*}{d\mu_{A,j}} = \underbrace{\frac{\partial \ln p_h}{\partial p_j} \frac{dp_j}{d\mu_{A,j}}}_{\text{aggregate demand spillover } (>0)} + \underbrace{\frac{\partial \ln p_h}{\partial L_h} \frac{dL_h^*}{d\mu_{A,j}}}_{(<0) \text{ (ambig.)}} \quad (12)$$

$$\frac{d\mathbb{E}[\ln w_h]^*}{d\mu_{\gamma,j}} = \underbrace{\frac{\partial \ln p_h}{\partial p_j} \frac{dp_j}{d\mu_{\gamma,j}}}_{\text{aggregate demand spillover } (>0)} + \underbrace{\frac{\partial \ln p_h}{\partial L_h} \frac{dL_h^*}{d\mu_{\gamma,j}}}_{(<0) \text{ (ambig.)}} \quad (13)$$

$$\frac{d\mathbb{E}[\ln w_h]^*}{d\sigma_{A,j}^2} = \underbrace{\sum_k \frac{\partial \ln p_h}{\partial p_k} \frac{\partial p_k}{\partial L_k} \frac{dL_k^*}{d\sigma_{A,j}^2}}_{\text{aggregate demand spillover } (<0)} + \underbrace{\frac{\partial \ln p_h}{\partial L_h} \frac{dL_h^*}{d\sigma_{A,j}^2}}_{(<0) \text{ } (>0)} < 0 \quad (14)$$

$$\frac{d\mathbb{E}[\ln w_h]^*}{d\sigma_{\gamma,j}^2} = \underbrace{\sum_k \frac{\partial \ln p_h}{\partial p_k} \frac{\partial p_k}{\partial L_k} \frac{dL_k^*}{d\sigma_{\gamma,j}^2}}_{\text{aggregate demand spillover } (<0)} + \underbrace{\frac{\partial \ln p_h}{\partial L_h} \frac{dL_h^*}{d\sigma_{\gamma,j}^2}}_{(<0) \text{ } (>0)} < 0 \quad (15)$$

Intuition: The wage impact on other careers h operates through two distinct general equilibrium channels. First, the **labor reallocation effect**: if an overwhelming share of individuals rushes into safer, more appealing paths, the emerging surplus of workers depresses their expected wages. Conversely, the paths they abandon experience a labor shortage, driving up their wages. Second, the **aggregate demand spillover**: because the goods produced by different careers are all combined to satisfy aggregate demand, a shock to sector j acts as a demand shift for sector h . A positive productivity shock in j boosts overall economic demand, making the economy richer and raising the demand (and wages) for all other careers. Conversely, an increase in uncertainty in j forces workers to flee to less productive but safer paths. This inefficiency shrinks aggregate demand, creating a negative demand spillover that drags down wages across the entire economy.

3 Measuring the AI Shocks

To implement the model quantitatively, we need to measure the expected values ($\mu_{A,j}$ and $\mu_{\gamma,j}$) and the uncertainties ($\sigma_{A,j}^2$ and $\sigma_{\gamma,j}^2$) of the automation and augmentation shocks for each training path j . I compute these moments by categorizing the tasks comprising each training path based on their current and potential AI exposure.

This categorization relies on two main sources: the actual task penetration of AI (Massenkoff and McCrory, 2026), and the theoretical automatability and augmentation potential of tasks (Althoff and Reichardt, 2026). The task penetration is zero if, at the moment, AI is not used in that task. If instead AI is already used in that task, its penetration is greater or equal to 1/2, with values closer to 1 meaning that AI is more

automating than augmenting that task (see Massenkoff and McCrory (2026) for more details). Theoretical automatability is either 1 if the task is potentially automatable or 0 otherwise. Theoretical augmentation is 1 if the task is not augmentable and greater than 1 if it is potentially augmentable, for example a value of 1.1 means that AI can boost labor productivity by 10% in that task (see Althoff and Reichardt (2026) for more details).

First, I define the *current status* of a task:

- **Automated:** The actual task penetration is greater than or equal to 0.9 and the task is automatable (automatable = 1).
- **Augmented:** The actual task penetration is greater than or equal to $1/2$, the task is not automatable (automatable = 0). Alternatively, the task is automatable (automatable = 1), but the current task penetration is less than 0.9.
- **Neither:** If the actual task penetration is zero, the task is currently neither automated nor augmented.

Second, I define the *potential status* of a task:

- If a task is already automated, its potential status is automated (with no uncertainty).
- If a task is already augmented and it is not automatable, its potential status is augmented (with no uncertainty).
- **Potentially automatable:** The task’s actual penetration is lower than 0.9, but the task is theoretically automatable (automatable = 1).
- **Potentially augmentable:** The task’s actual penetration is zero, it is not automatable (automatable = 0), and its augmentation factor is greater than 1.
- **Not affected by AI:** The task’s actual penetration is zero, it is not automatable (automatable = 0), and its augmentation factor is exactly 1.

To formalize this aggregation, let $\mathcal{T}_j$ be the set of tasks associated with training path j , and let $\theta_{j,\tau}$ denote the importance weight of task τ (such that $\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} = 1$). I treat the ultimate AI exposure of each task as a random variable. Let $a_\tau \in \{0, 1\}$ be an indicator of whether task τ will eventually be automated, and let $b_\tau \in \{0, 1\}$ be an indicator of whether it will be augmented.

For automation, if a task is already automated, its future outcome is certain: $\mathbb{E}[a_\tau] = 1$ and $\text{Var}(a_\tau) = 0$. Conversely, if it is not even potentially automatable, $\mathbb{E}[a_\tau] = 0$ and $\text{Var}(a_\tau) = 0$. The uncertainty lies entirely with the tasks that are *potentially*

automatable but not yet automated. For these tasks in the “gap”, I assume a probability $p_A \in (0, 1)$ that the theoretical automatability translates into actual automation. Thus, for tasks in this gap, $\mathbb{E}[a_\tau] = p_A$ and $\text{Var}(a_\tau) = p_A(1 - p_A)$.

The expected automation share $\mu_{A,j}$ for training path j is the weighted sum of these task-level expectations:

$$\mu_{A,j} = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} \mathbb{E}[a_\tau] = S_{A,j}^{cur} + p_A (S_{A,j}^{pot} - S_{A,j}^{cur}) \quad (16)$$

where $S_{A,j}^{cur}$ is the currently automated (weighted) share and $S_{A,j}^{pot}$ is the maximum potentially automatable (weighted) share. To formalize the possibility that AI success is correlated across tasks within the same training path, I introduce a correlation parameter $\kappa \in [0, 1]$. A value of $\kappa = 0$ implies independent task automation (maximizing diversification), whereas $\kappa = 1$ implies perfect correlation (AI automates either all potential tasks or none). The variance of the automation shock then decomposes into a convex combination of the independent and perfectly correlated variances:

$$\sigma_{A,j}^2 = p_A(1 - p_A) \left[(1 - \kappa) \sum_{\tau \in \text{Gap}_{A,j}} \theta_{j,\tau}^2 + \kappa \left(\sum_{\tau \in \text{Gap}_{A,j}} \theta_{j,\tau} \right)^2 \right] \quad (17)$$

where $\text{Gap}_{A,j}$ is the set of tasks that are potentially automatable but not yet automated. First, notice that the total variance is strictly increasing in the total automation gap $\sum_{\tau \in \text{Gap}_{A,j}} \theta_{j,\tau}$; therefore, training paths with a higher unrealized potential for automation inherently face higher uncertainty. Second, the term $\sum \theta_{j,\tau}^2$ acts as a Herfindahl-Hirschman Index (HHI) of task concentration. This implies that, especially when $\kappa < 1$, holding the gap share constant, training paths that are highly specialized in a few uncertain tasks face significantly more risk than paths with a diversified portfolio of many small tasks.

A symmetric logic applies to augmentation. Let g_τ be the theoretical augmentation factor for task τ (where $g_\tau > 1$ means the task is augmentable, and $g_\tau = 1$ means no gain). For tasks that are strictly augmented (already augmented and *not* potentially automatable), their outcome is certain: $\mathbb{E}[b_\tau] = 1$ and $\text{Var}(b_\tau) = 0$. Conversely, if the task is not even potentially augmentable and not currently augmented, $\mathbb{E}[b_\tau] = 0$ and $\text{Var}(b_\tau) = 0$.

The uncertainty in augmentation, captured by the set $\text{Gap}_{\gamma,j}$, arises from two distinct sources. First, tasks that are *potentially augmentable* but not yet augmented have a probability p_γ of realizing this productivity boost. Second, tasks that are *already augmented* but also *potentially automatable* face the risk of full automation; if automation succeeds (with probability p_A), the human is replaced and the augmentation factor

effectively drops back to 1. Thus, they only have a probability $1 - p_A$ of remaining augmented. Assuming for simplicity that $p_\gamma = 1 - p_A \equiv p$ (as implemented in the baseline calibration where both probabilities are 0.5), both sources of uncertainty yield an expected indicator $\mathbb{E}[b_\tau] = p$ and a variance $Var(b_\tau) = p(1 - p)$. The expected augmentation factor and its variance for training path j are then:

$$\mu_{\gamma,j} = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau} (1 + \mathbb{E}[b_\tau](g_\tau - 1)) \quad (18)$$

$$\sigma_{\gamma,j}^2 = p_\gamma(1 - p_\gamma) \left[(1 - \kappa) \sum_{\tau \in \text{Gap}_{\gamma,j}} \theta_{j,\tau}^2 (g_\tau - 1)^2 + \kappa \left(\sum_{\tau \in \text{Gap}_{\gamma,j}} \theta_{j,\tau} (g_\tau - 1) \right)^2 \right] \quad (19)$$

Just as with automation, this variance is driven by two key factors: the size of the potential augmentation gap (meaning paths with a higher potential productivity boost naturally face higher uncertainty) and the concentration of that potential across a few specific tasks.

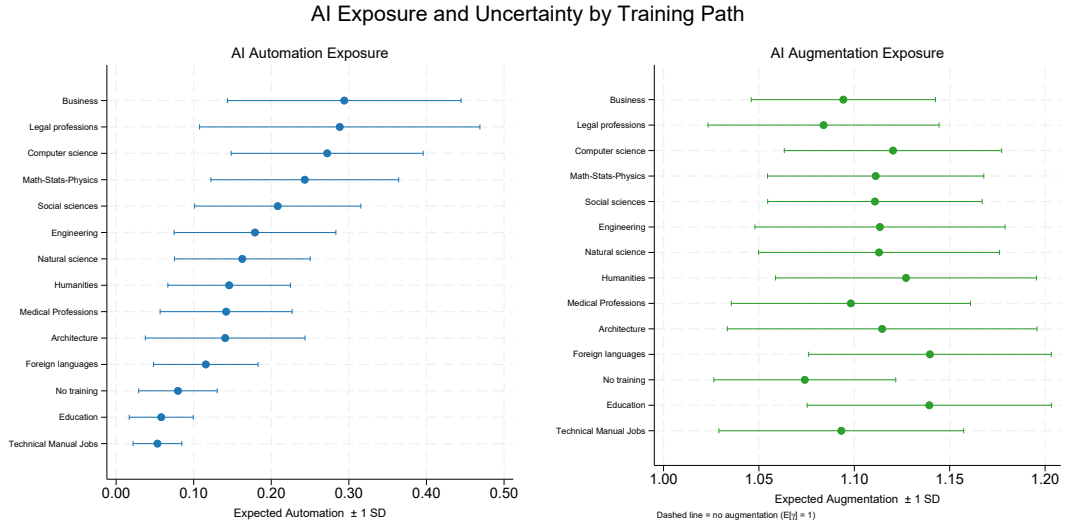


Figure 1: Expected Automation and Augmentation across Training Paths (with 1 standard deviation confidence interval).

Figure 1 displays the expected automation ($\mu_{A,j}$) and expected augmentation ($\mu_{\gamma,j}$) shocks across different training paths. Furthermore, the confidence intervals capture differences in uncertainty across training paths. For instance, the training paths with the most automation uncertainty are legal professions and business, because many of their tasks are potentially automatable, but not yet automated. Computer science has a lower uncertainty compared to them (still relatively high) because some of its tasks have already been automated. On the other hand, paths like technical manual jobs and education have a very low expected automation (and uncertainty) because their tasks

are more difficult to automate. As far as augmentation is concerned, both expectations and uncertainty are similar across training paths; as a result, its role in shaping labor reallocation is limited compared to automation.

4 Calibration

The quantitative implementation of the model proceeds in two main stages. First, I calibrate the model to a pre-AI baseline equilibrium (representing the year 2022) where AI shocks are not yet anticipated. This allows me to recover the fundamental structural parameters of the economy, particularly the unobserved training costs. Second, I introduce the measured AI shocks (both expected productivity and uncertainty) derived in Section 3 to simulate the new post-AI equilibrium and decompose its effects on employment and wages.

4.1 Pre-AI Baseline Equilibrium (2022)

In the pre-AI steady state, agents do not anticipate the arrival of Artificial Intelligence. Consequently, their expectations about future automation and augmentation are zero ($\mu_{A,j} = \mu_{\gamma,j} = 0$), and there is no uncertainty regarding these shocks ($\sigma_{A,j}^2 = \sigma_{\gamma,j}^2 = 0$). Without uncertainty, the risk premium is zero, and the certainty equivalent Ω_j collapses exactly to the expected current log-wage: $\Omega_j^{pre} = \ln w_j^{2022}$.

Using the optimal labor supply derived in Equation (4), the employment share for training path j in the pre-AI equilibrium is given by:

$$L_j^{2022} = \frac{\exp((\beta \ln w_j^{2022} - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^J \exp((\beta \ln w_h^{2022} - \bar{C}_h^*)/\zeta)} \quad (20)$$

where L_j^{2022} and w_j^{2022} are the observed employment shares and median wages from the 2022 American Community Survey (ACS) and Bureau of Labor Statistics (BLS) data, across the 14 training paths.

By observing the actual labor allocations and wages, I can invert this mapping to exactly recover the unobserved normalized training costs $\bar{C}_j^*$. Normalizing the cost of the no-training option ($j = 0$) to zero ($\bar{C}_0^* = 0$), the structural cost parameters are identified as:

$$\bar{C}_j^* = \beta(\ln w_j^{2022} - \ln w_0^{2022}) - \zeta(\ln L_j^{2022} - \ln L_0^{2022}) \quad (21)$$

On the labor demand side, following the procedure outlined in Althoff and Reichardt (2026), I assume that occupational goods are substituted with a constant elasticity of substitution (CES) ϵ , so that demand for occupation j satisfies $D_j(\mathbf{p}) \propto \alpha_j p_j^{-\epsilon}$. I

set the elasticity of substitution between occupations $\epsilon = 1.57$ —the midpoint between 1.81 (Burststein et al., 2019) and 1.34 (Caunedo et al., 2023). The CES importance weights α_j are then identified from observed occupational wage shares and the estimated occupational prices because for any two occupations i and j :

$$\frac{\alpha_i}{\alpha_j} = \left(\frac{p_i}{p_j}\right)^{\epsilon-1} \times \frac{\text{Wage share of occupation } i}{\text{Wage share of occupation } j} \quad (22)$$

Because no tasks are automated in the pre-AI baseline economy, the labor share is exactly one ($\Gamma_j = 1$) and output per worker is normalized to one ($Y_j = 1$). Consequently, the competitive wage equation $w_j = p_j Y_j \Gamma_j$ simplifies to $p_j = w_j$. Thus, I can directly substitute the observed 2022 wages w_j^{2022} for the prices p_j in the equation above to perfectly calibrate the demand shifters α_j .

4.2 Externally Calibrated Parameters

To simulate the model quantitatively, I calibrate several fundamental parameters directly from the recent literature on task-based models, relying primarily on the estimates provided by Althoff and Reichardt (2026).

Macro Elasticities and Preferences: I set the elasticity of substitution across tasks within an occupation to $\rho = 0.49$, implying that tasks are complementary. For the agents’ preferences, the annual discount factor is set to $\beta = 0.813$, corresponding to an annual discount rate of 3% and considering 7 years (from 18 to 25) for the training period, and the coefficient of relative risk aversion is set to a standard macro value of $\eta = 2$. Finally, following Althoff and Reichardt (2026), the Gumbel dispersion parameter for idiosyncratic occupational preferences is set to $\zeta = 0.053$.

Task Exposure and Augmentation: The expected shocks (μ) and uncertainties (σ^2) defined in Section 3 require calibrating the probability that a task in the “gap” is eventually exposed to AI. To reflect a state of maximum agnostic uncertainty regarding the future of these tasks, I set the adoption probabilities to $p_A = 0.5$ and $p_\gamma = 0.5$. Furthermore, I use the task-specific augmentation factors g_τ from (Althoff and Reichardt, 2026).

Capital-Labor Substitution: While the pre-AI baseline economy assumes no tasks are automated by generative AI (so the initial labor share is exactly $\Gamma_j^{pre} = 1$), simulating the post-AI economy requires quantifying how the labor share responds to the automation shock. Rather than calibrating task-specific capital rental rates and productivities, I follow Althoff and Reichardt (2026) and assume that the cost savings of using AI instead of labor for automatable tasks is 27%, yielding a relative cost of capital $\chi = 0.73$. Using their structural derivation, the post-AI labor share for any

training path j can be approximated as:

$$\Gamma_j^{post} \approx \frac{\chi^{\rho-1} (1 - \mu_{A,j})}{\mu_{A,j} + \chi^{\rho-1} (1 - \mu_{A,j})} \quad (23)$$

where $\mu_{A,j}$ is the expected total share of tasks automated by AI in training path j (as derived in Section 3).

4.3 Simulating the Post-AI Equilibrium

Once the structural parameters are pinned down, I introduce the AI shocks. For each training path j , I plug in the expected values $(\mu_{A,j}, \mu_{\gamma,j})$ and variances $(\sigma_{A,j}^2, \sigma_{\gamma,j}^2)$ computed in Section 3.

These shocks alter the certainty equivalent Ω_j^{post} for each training path through both the direct expected productivity shifts and the endogenous risk premiums. The new post-AI equilibrium is defined as the fixed point where:

1. Risk-averse agents optimally reallocate across training paths based on the new certainty equivalents, forming the post-AI labor supply L_j^{post} .
2. The new labor supply alters the production of occupational goods, which in turn adjusts the equilibrium prices p_j^{post} according to the CES demand function.
3. Expected log-wages $\mathbb{E}[\ln w_j]^{post}$ adjust endogenously to clear the market.

By comparing the post-AI equilibrium to the pre-AI baseline, we can quantify the total effect of AI on employment and expected wages, and further decompose these effects into the demand channel, the uncertainty channel, and the crowding effect.

5 Results and Decomposition

This section presents the quantitative predictions of the calibrated model. To isolate the contribution of each economic mechanism, the full post-AI equilibrium is decomposed sequentially into four channels: the *expected automation* effect, the *expected augmentation* effect, and the *uncertainty* effects associated with both automation and augmentation shocks.

5.1 Employment Reallocation and the Flight to Safety

Figure 2 presents the decomposition of AI-induced employment shifts across the 14 training paths.

The *expected automation* channel drives the primary structural transformation. As AI substitutes for cognitive tasks, labor demand shifts away from highly automatable cognitive paths such as business, computer science, and legal professions. Conversely, paths with high physical or interpersonal requirements—such as technical manual jobs and education, as well as the “no training” residual category, experience positive labor inflows. The *expected augmentation* channel plays a comparatively minor role in employment reallocation, as human-enhancing technologies boost output across most sectors relatively symmetrically without dramatically shifting relative labor demand.

Crucially, the *uncertainty* over automation triggers a massive secondary reallocation. Risk-averse agents actively flee the highly uncertain training paths (e.g., business and legal professions). This flight from uncertainty strongly amplifies the initial expected automation shock, accelerating the collapse of employment in exposed careers and crowding workers into the perceived safer manual jobs. Uncertainty regarding augmentation has virtually zero impact on employment, again because the augmentation uncertainty is relatively similar across training paths.

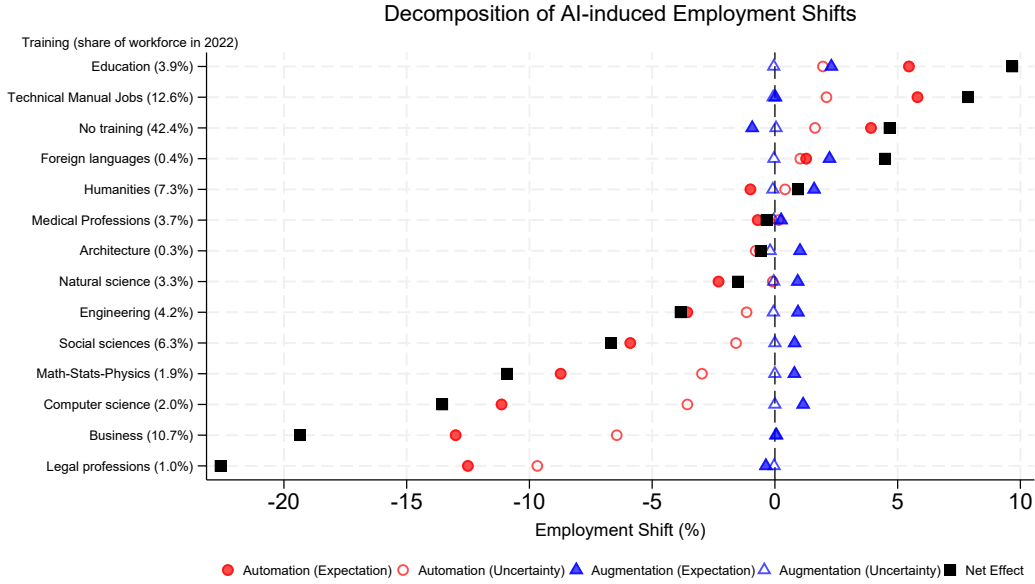


Figure 2: Decomposition of AI-induced employment shifts across training paths. The total net effect (black squares) is decomposed into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles). Shifts are expressed as percentage changes relative to the pre-AI baseline share.

5.2 Wage Effects and the Endogenous Risk Premium

The wage decomposition (Figure 3) reveals a starkly different pattern, underscoring the power of general equilibrium forces. The expected AI shocks—both automation and augmentation—are unequivocally beneficial for aggregate productivity, generating an across-the-board increase in expected wages ranging from +4.2% to +5.4% from automation, and +9.3% to +9.4% from augmentation.

However, the employment reallocation generated by the uncertainty channel creates a *crowding effect* that fundamentally reshapes relative wages. Training paths that are relatively safer (such as technical manual jobs) experience downward wage pressure as labor congestion erodes marginal productivity, costing them between 1.2% and 1.9% in expected wages.

In sharp contrast, the training paths suffering the largest flight of workers—such as legal professions, business, and computer science—experience a labor scarcity effect. This scarcity pushes expected wages significantly upward, generating an *endogenous risk premium* that compensates the remaining workers. For instance, legal professions see an additional +6.0% wage premium purely due to the flight of risk-averse agents. Consequently, these highly exposed cognitive paths ultimately experience the largest total net wage gains (up to +19.6%).

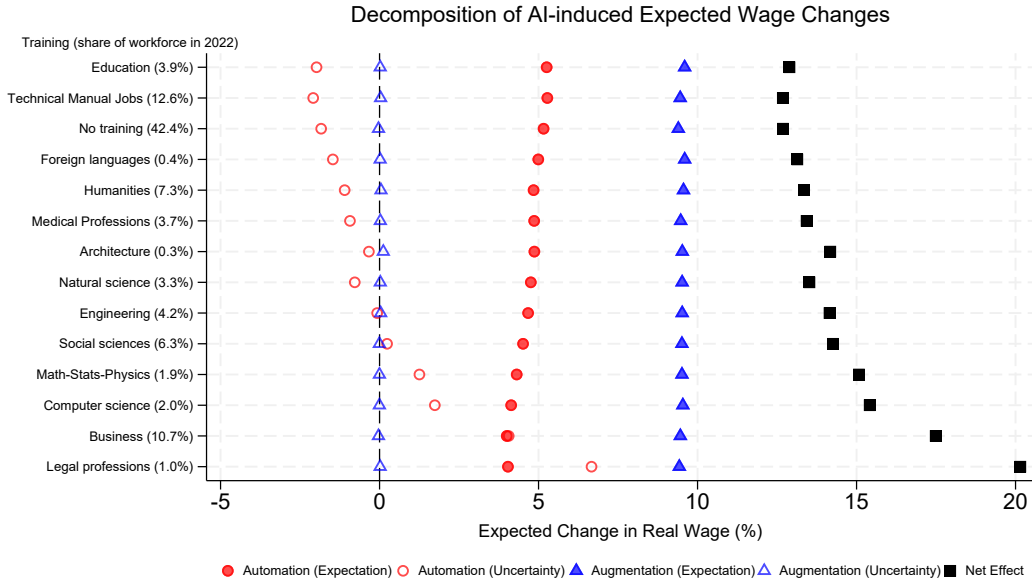


Figure 3: Decomposition of AI-induced expected wage changes across training paths. The total net wage change (black squares) is decomposed into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles).

To further highlight this endogenous risk premium, Figure 4 plots the expected wage changes against the variance of the AI shock across the different training paths. The evident positive relationship confirms that paths facing higher technological uncertainty compensate workers with higher expected wages. This highlights the general equilibrium effect: as risk-averse workers flee highly volatile training paths, the resulting labor scarcity mechanically drives up the expected wages in those sectors.



Figure 4: Expected Wage Changes vs. AI Shock Variance across Training Paths.

6 Conclusion

This paper investigates how Artificial Intelligence reshapes human capital investment decisions when workers face profound technological uncertainty. To study this, I develop a structural task-based choice model in which risk-averse agents choose among different training paths, evaluating both the expected demand shifts and the variance surrounding AI's future impact. By calibrating the model to the US labor market using ACS and BLS data, along with task-level AI exposure and penetration metrics, I decompose the post-AI labor market equilibrium into expected technology effects and risk-driven reallocation channels.

The theoretical and quantitative findings highlight three main insights. First, expected automation drives a structural shift of labor away from the most exposed cognitive professions and toward manual roles. In contrast, expected augmentation and its associated uncertainty play a limited role in employment reallocation, as their expected effects are largely uniform across training paths. Second, uncertainty about

future automation is a major determinant of career choices: risk-averse individuals systematically flee highly uncertain cognitive paths. Third, this flight to safety creates powerful general equilibrium effects. If we ignore automation uncertainty, AI’s impact on expected wages is highly similar across all training paths. However, risk-driven reallocation fundamentally reshapes relative wages: the crowding of workers into perceived safe havens puts downward pressure on their wages, whereas the labor scarcity in exposed cognitive sectors generates an endogenous risk premium that drives their expected wages up significantly.

Overall, the results demonstrate that traditional employment forecasts focusing solely on expected point estimates of AI’s impact overlook a critical mechanism. The uncertainty surrounding Artificial Intelligence is not just a statistical artifact—it is a primary driver of labor supply that actively reshapes the distribution of human capital and the structure of future wages.

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